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8 November 1982
SUPERSEDING
MIL-P-23857A (DOCKS)
28 July 1964

MILITARY SPECIFICATION

PUMPING ASSEMBLIES, ASPHALT, BULK TRANSFER

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers two sizes of pumping assemblies designed for bulk transfer of heated asphalt in a mobile, continuous-mix asphalt plant. The pumping assemblies consist of a jacketed, positive-displacement, rotary pump; an electric motor with magnetic starter and push-button station; a jacketed strainer; drive mechanism; and other components and accessories specified herein. The assemblies shall be mounted on a steel skid-base to form an integrated, portable, asphalt pumping unit.

1.2 Classification. Pumping assemblies shall be of the following sizes, as specified (see 6.2.1).

Size 1 - 90 gallons per minute (gpm).
Size 2 - 200 gpm.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2.1), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4320

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SPECIFICATIONS

FEDERAL

TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

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MIL-P-116 - Preservation, Methods of.
MIL-V-173 - Varnish, Moisture and Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-C-3774 - Crates, Wood, Open, 12,000- and 16,000- Pound Capacity.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts, Packaging of.
MIL-U-18182 - Union, Pipe, Forged Steel, Threaded, Water-Oil-Gas (2,000, 3,000 and 6,000 lb/in^2).
MIL-B-26195 - Boxes, Wood-cleated, Skidded, Load-Bearing Base.
MIL-P-46093 - Primer Coating, Synthetic (For Brake Drums).
MIL-C-52950 - Crates, Wood, Open and Covered.

STANDARD

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B2.1 - Pipe Threads (Except Dryseal).
B15.1 - Safety Standard for Mechanical Power Transmission Apparatus.
B16.9 - Factory-Made Wrought Steel Buttwelding Fittings.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A278 - Gray Iron Castings for Pressure-Containing Parts for Temperatures Up to 650°F (345°C), Specification for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19102.)

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HYDRAULIC INSTITUTE (HI)

Standards of Hydraulic Institute

(Application for copies should be addressed to the Hydraulic Institute, 14600 Detroit Ave., Suite 712, North Cleveland, OH 44107.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS - 1 - General Standards for Industrial Controls and Systems.
MG - 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS.

3.1 Pumping assembly data sheet. Unless otherwise specified (see 6.2.1), each bidder shall submit, with his bid proposal, a completed form in accordance with the Pumping Assembly Data Sheet shown in figure 1. Information furnished shall reflect the actual characteristics of the components to be furnished. The characteristics shall be equal to or exceed the minimum requirements specified herein.

3.2 First article. When specified (see 6.2.1), the contractor shall furnish a pumping assembly for first article inspection and approval (see 4.2.1 and 6.3).

3.3 Standard commercial product. The pumping assembly shall, as a minimum, be in accordance with the requirements of this specification and shall be the

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manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product shall be included in the pumping assembly being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.6 Design. The pumping assembly shall be designed for the bulk transfer of heated asphalt (750 to 1,500 Saybolt Seconds Universal (SSU)) from asphalt melters to storage tanks. Pressure containing parts of the pump and strainer, including hot oil jackets, shall be designed for an allowable working pressure of not less than 100 pounds per square inch gage (psig) at a temperature of 350° Fahrenheit (F) (see 4.4.1). Except as specified herein, the details of the design of individual components, including clearances for rotating parts in the pump casing, shall be in accordance with the recommendations of the component's manufacturer for the specified design and operating conditions. The pumping assembly shall be so designed that no permanent mechanical damage will result should the pump be operated with the viscosity of the asphalt up to 20,000 SSU. The design of the assembly shall permit accessibility for maintenance and service in the field.

3.7 Pump. The pump shall be a heavy-duty positive displacement, rotary pump. The pump casing and stuffing box shall be jacketed to permit transfer of heat from hot oil to the asphalt and packing. The pump assembly shall be equipped with a relief valve to prevent buildup of excessive pressure in the discharge line. The relief valve shall be designed to bypass asphalt from the discharge back to the suction side of the pump. Unless otherwise specified (see 6.2.1), the valve shall be set to relieve at 50 psig. The nominal pipe size (nps) of the pump suction and discharge connections shall be not less than 2.5 inches for size 1 and 3 inches for size 2. The direction of pump rotation shall be conspicuously and permanently marked on the pump casing.

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3.7.1 Capacity. The pump shall be capable of delivering heated asphalt having a viscosity of 1,000 ssu against a total discharge head of 25 psig at the rate of not less than 90 gpm for size 1 pumps and not less than 200 gpm for size 2 pumps. The pump speed at the rated capacity shall be the manufacturer's recommended speed for the conditions specified herein. Pumps requiring a speed in excess of 450 revolutions per minute to provide the specified capacity will not be acceptable unless approved in writing by the contracting officer.

3.7.2 Construction. The pump shall be of the sliding vane type or external or internal gear type, as defined in the Standards of the Hydraulic Institute. The pump casing shall be cast iron having properties equivalent to or better than class 25 of ASTM A278. The shaft shall be steel, hardened and ground, in accordance with the manufacturer's standard practice. The gears shall be steel, hardened and ground, or cast iron finished in accordance with the manufacturer's standard practice. Leakage along the pump shaft shall be controlled by packing suitable for the specified design temperatures and use. The pump shall be furnished with an extended shaft which shall be supported by three bearings. Two bearings, one on each side of the casing, shall be bronze, sleeve type or antifriction type. The end of the pump shaft shall be supported, adjacent to the sheave, by a self-alining, heavy-duty, pillow block bearing or by an outboard, antifriction bearing mounted integrally with an extension of the pump casing. The pillow block or outboard bearing shall be equipped with antifriction bearings designed for a rated life expectancy of not less than 2,000 hours based on the intended application. The casing and stuffing box jackets shall each be furnished with not less than two tappings for connection of the hot oil tubing (see 3.9.2). The stuffing box connection and the casing connection shall be not less than 0.75 inch nps.

3.8 Strainer. The strainer shall consist of a cast iron or steel body with cover, a straining mechanism, and a hot oil jacket. The nps of the strainer inlet and outlet connections shall be not less than 2.5 inches for size 1 and 3 inches for size 2. The cover shall be bolted to the body to permit removal for cleaning and replacement of the straining mechanism. A drain shall be provided in the base of the strainer for flushing and cleaning the sump. The straining mechanism shall be made of wire or perforated metal and shall provide a degree of filtration equal to or better than number 3 mesh. The jacket chamber shall be provided with not less than two threaded openings for connection of the hot oil jumper tubing (see 3.9.2). The hot oil connection shall be not less than 0.75 inch nps. The strainer shall be of sufficient size to assure that the pressure drop, when operating under the conditions and capacity specified (see 3.7.1), will not exceed the difference between the pump manufacturer's allowable suction lift and a system suction lift of 3 psig external to the strainer.

3.9 Piping and connections.

3.9.1 Asphalt system. The inlet and outlet connections of both the pump and strainer shall be flanged. The connection between the strainer outlet and pump inlet shall be complete with all necessary elbows, reducers, and couplings. All elbows, reducers, couplings, and flanges shall be connected by

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welding. Dimensions of all elbows, reducers, and couplings shall be in accordance with ANSI B16.9. Both the inlet connection of the strainer and the outlet connection of the pump shall be provided with an adapter consisting of a bolted flange having a threaded nipple welded to the flange. The threaded nipple shall be 4 inches by 2.5 inches for size 1 and 4 inches by 3 inches for size 2. Flanged connections shall be provided with gaskets suitable for the specified operating conditions.

3.9.2 Hot oil system. Connections shall be provided to permit circulation of hot oil from an external source, through the pump jacket to the strainer jacket, through the strainer jacket to the stuffing box, and back to the external source. Jumper connections between the pump jackets and the strainer jacket shall consist of flexible metal tubing having a 1-inch nominal inside diameter. Tubing shall be rated at not less than 200 psig at 350°F and shall be constructed of corrugated steel with a corrosion-resistant wire-braid covering. Each end of the jumpers shall terminate in a threaded male fitting. Jumpers shall be attached to jacket connections by forged steel unions in accordance with MIL-U-18182. Unions shall be class 1 or better and may be coated or uncoated. The nps of the unions shall be 1 inch with suitable elbows, nipples, and reducers furnished, as required, for connection to the appropriate tappings in the jackets. Two additional 1-inch unions, conforming to requirements herein, shall be furnished and installed in the hot oil inlet and outlet connections to provide connection in the field to the hot oil system.

3.9.3 Discharge hose. One 20-foot length of flexible metal discharge hose shall be furnished with each pumping assembly. The hose shall be fabricated of corrugated steel with a corrosion-resistant wire-braid covering. The hose shall be capable of withstanding a pressure not less than 150 psig at 450°F and shall be equipped with male pipe threads at each end. One union, suitable for connecting the hose to the pump discharge adapter, shall be furnished. The union shall be malleable iron or steel. The coupling nut shall be furnished with two 180° opposed lugs (hammerbar or handlebar type) to facilitate quick-make and -break connection of the hose to the pump. The nominal size of the hose, male threaded ends, and union shall be 2.5 inches for size 1 and 3 inches for size 2.

3.9.4 Threaded connections. All threaded connections in the asphalt system, hot oil system, and discharge hose shall conform to ANSI B2.1, NPT designation.

3.10 Electrical components.

3.10.1 Motor. The motor shall have an explosion proof enclosure and be constructed, classified, and rated in accordance with NEMA MG-1. The motor shall be designed to operate on a 208-volt, 3-phase, 60-Hertz (Hz) power supply. The rpm and horsepower rating shall be sufficient to operate the pump under the specified conditions.

3.10.2 Magnetic starter. A magnetic starter, conforming to the applicable requirements of NEMA ICS-1, shall be furnished and mounted as specified in 3.12. The starter shall be furnished with a weatherproof enclosure. The

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starter shall provide thermal overload and undervoltage protection for the motor. Thermal overload heaters of appropriate rating for the motor current draw shall be furnished and installed. The pushbutton stop-and-start station, the magnetic starter, and the motor shall be connected by wiring and installed in accordance with NFPA No. 70, as applicable to outside installations.

3.11 Drive. The pump shall be connected to the electric motor by a direct-coupled reduction gear or by a multiple V-belt drive. Reduction gears shall be machine-cut and run in oil in a sealed weatherproof case. V-belts shall be industrial standard belts. The drive mechanism shall be in accordance with the manufacturer's standard practice for the specified conditions. All rotating elements shall be enclosed by a safety guard. Safety guards shall be attached to the skid and shall be removable to permit access for repairs and replacements. The guard shall be in accordance with ANSI B15.1.

3.12 Skid base. The pumping assembly shall be mounted on a skid base consisting of a bedplate, skids, and required mountings for individual components. The bedplate shall be carbon steel plate having a nominal thickness of 0.25 inch. The skids shall be suitable structural steel shapes which will provide a clearance of not less than 4 inches under the bedplate and a bearing width per skid of not less than 3 inches. The skids shall be welded on the outside edge of the longest dimension of the bedplate and shall extend between 4 and 6 inches beyond the edges of the bedplate perpendicular to the skid. The ends of each skid shall be rounded with a radius equivalent to the depth of the skid. A hole not less than 2 inches in diameter shall be cut in the web near both ends of each skid so that the pump assembly can be lifted with slings. Individual pedestal mountings shall be furnished for the motor, pump, pillow block, and other components, as required, to provide vertical alinement of components and adequate clearance of the pump sheave over the bedplate. Mounting holes for components shall be slotted, as required, to permit horizontal alinement, maintenance, and replacement of components. The motor starter and pushbutton station shall be mounted vertically on a rigidly braced stand constructed of suitable structural steel shapes and plate. The height of the controller stand shall be sufficient to provide clearance for removal of the magnetic starter cover. Mounting of electrical components shall be such as to insure electrical continuity to ground. The overall dimensions of the bedplate shall be not greater than 46 inches by 52 inches for size 1 and 48 inches by 62 inches for size 2. No part of any component shall extend beyond the edge of the bedplate.

3.13 Lubrication. Unless otherwise specified (see 6.2.1), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be lubricated as specified in section 5. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment (1,000 pounds per square inch or higher) will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.14 Identification plate. An identification plate will be furnished by the contracting officer for each pumping assembly. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose,

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and securely affix a plate to each pumping assembly in a conspicuous place with brass screws or bolts not less than 0.125 inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.15 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2.1). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.16 Fungus resistance. When specified (see 6.2.1), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.17 Workmanship. The quality of workmanship shall be in accordance with the requirements specified herein. The pumping assembly shall be free of defects which will adversely affect the life, serviceability, or appearance of the unit.

3.17.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.17.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.17.3 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or

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failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one pumping assembly unit when a first article is required (see 3.2 and 6.2.1). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4, and the packaging inspection of 4.5.

4.3 Examination. Each pumping assembly shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests.

4.4.1 Hydrostatic test. The asphalt system and the hot oil system of the completely assembled pumping unit shall be hydrostatically tested at a pressure

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equal to one and one-half times the allowable working pressure specified in 3.6. Each system shall be tested separately. Hydrostatic tests shall be made using preservatives equal to type P-10 as specified in MIL-P-116 and shall be for not less than 10 minutes duration.

4.4.2 Performance test. The pump shall be tested to determine compliance with the requirements of 3.7.1. Tests shall be conducted and reported in accordance with the Rotary Pump Section of the Standards of the Hydraulic Institute. The test medium shall be a clean petroleum oil of approximately 250 ssu viscosity at a pumping temperature between 60°F. and 130°F. Test results shall be corrected as required to reflect the difference in viscosity of the test medium and asphalt having the viscosity specified in 3.7.1.

4.4.2.1 Certification. When specified (see 6.2.1), certified test reports indicating that the model of pump being furnished has been tested in accordance with 4.4.2 and meets the requirements of 3.7.1 shall be acceptable in lieu of actual performance tests.

4.4.3 Functional test. The magnetic starter shall be connected to a 3-phase, 60-Hz, nominal 208-volt power source. With the drive mechanism removed, the electric motor shall be operated by actuation of the pushbutton control through at least three off and on cycles.

4.5 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or commercial as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Unit protection. The complete pumping assembly shall be preserved by method II. The submethod shall be at the contractor's option.

5.1.1.3 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws, and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.4 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed, and mating parts on the equipment, shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

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5.1.1.5 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.6 Unprotected surfaces. Unprotected, unpainted, and uncoated metal surfaces, including threaded surfaces, requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically specified herein shall be coated with P-1, P-2, P-6, or P-19.

5.1.1.7 Asphalt filter, pump, and oil lines. The interior surfaces of the asphalt filter, pump, and oil lines shall be preserved by flushing type P-10 preservative. Upon completion, the preservative shall be completely drained.

5.1.1.8 Electric motor and starter. The electric motor and starter require no preservation when the complete pumping assembly is preserved by method II.

5.1.1.9 Drive belts and pulleys. Drive belts shall be removed or released from tension. Removed belts shall be preserved by method IC-1 or IC-3. Unpainted faces shall be coated with primer conforming to TT-P-664 or MIL-P-46093. A weatherproof tag shall be attached in a conspicuous location indicating: "Belts have been (removed or released from tension). (Install or tension) prior to operation." Markings shall be applied to the tags with a waterproof material.

5.1.1.10 Discharge hose. The discharge hose requires no preservation when the complete pumping assembly is preserved by method II, and the discharge hose is contained within the method II barrier. When not contained within the method II barrier, the discharge hose shall be preserved as specified in 5.1.2.3.

5.1.1.11 Consolidation. Disassembled components for one complete pumping assembly shall be consolidated in one or more containers conforming to PPP-B-636, class weather-resistant. Contents shall be cushioned, blocked, and braced to prevent movement within the container. Containers shall be secured to the equipment within the method II barrier in such a manner to prevent movement and rupture of the barrier.

5.1.2 Level B. Level B preservation shall be the same as level A with the following exceptions and modifications:

5.1.2.1 Unit protection. The complete pumping assembly shall be preserved by method I.

5.1.2.2 Electric motor and starter. The electric motor and starter shall be preserved in accordance with the alternate level A requirements of MIL-E-16298.

5.1.2.3 Discharge hose. Surfaces of the hose requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated inside and outside with type P-10, grade 10 preservative. The hose shall be packed in a container conforming to PPP-B-636, class weather-resistant.

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5.1.3 Commercial. The equipment shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2.1).

5.2.1 Level A. The complete pumping assembly shall be packed in containers conforming to PPP-B-601, overseas type; PPP-B-621, class 2; or MIL-B-26195, type II with a plywood superstructure. Contents shall be cushioned, blocked, and braced to prevent movement within the container.

5.2.2 Level B. The complete pumping assembly shall be packed in domestic containers specified in 5.2.1 or in open crates conforming to MIL-C-3774, type II or MIL-C-52950, type V, style A or B with the alternate method of demountability. Contents in open crates shall be protected with a shroud as specified in the appendix to the applicable crate specification. Contents shall be cushioned, blocked, and braced to prevent movement within the container.

5.2.3 Commercial. The equipment shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The pumping assemblies covered by this specification are intended primarily for use in continuous-mix, heavy-duty mobile asphalt plants designed for Advanced Base applications. The assemblies are required to transfer heated asphalt from the barrel melters to the storage tanks.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Size of pumping assembly required (see 1.2).
c. Issue of document required, if other than as specified (see 2.1.1).
d. When a specification sheet is not required (see 3.1).
e. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.3).
f. When pump relief valve setting shall be other than 50 psig (see 3.7).
g. When means for lubrication shall be other than manufacturer's standard practice (see 3.13).
h. Color for finish coat (see 3.15).
i. When treatment for fungus resistance is required (see 3.16).
j. When certification is acceptable in lieu of performance tests (see 4.4.2.1).
k. Level of preservation and level of packing required (see 5.1 and 5.2).

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirements title	Applicable DID No.	Option
4.4.3.1	Certificate of Compliance	DI-E-2121	-----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete pumping assembly. The contracting officer should include specific instructions in all acquisition instruments regarding arrangement for examinations, tests, and approval of the first article.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:

Navy - YD

Project No. 4320-N197

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1. Date: _____ IFB NO.: _____
2. Bidder: _____
3. Address: _____
4. Pump
   a. Manufacturer: _____
   b. Manufacturer's model designation: _____
   c. Performance: _____gpm of asphalt at a viscosity of _____ssu against a total discharge head of _____psig at a rated speed of _____ rpm.
   d. Operating pressures (maximum):
      (1) Pump casing: _____ psig at _____ °F.
      (2) Jacket: _____ psig at _____ °F.
5. Strainer:
   a. Manufacturer: _____
   b. Manufacturer's model designation: _____
   c. Performance: _____ gpm of asphalt at a viscosity of _____ ssu with a pressure drop across the strainer of _____ psig.
   d. Operating pressures (maximum):
      (1) Asphalt side: _____ psig at _____ °F.
      (2) Jacket: _____ psig at _____ °F.
6. Motor:
   a. Manufacturer: _____
   b. Manufacturer's designation: _____
   c. Continuous horsepower rating: _____
   d. Speed at rated load: _____
   e. Nominal voltage: _____
   f. Type of enclosure: _____
   g. Temperature rise: _____
   h. Full-load amperes: _____
7. Magnetic motor starter:
   a. Manufacturer: _____
   b. Manufacturer's model designation: _____
   c. Continuous current rating: _____
8. Reduction gears (when applicable):
   a. Manufacturer: _____
   b. Manufacturer's designation: _____
   c. Gear ratio: _____
9. Belts (when applicable):
   a. Manufacturer: _____
   b. Manufacturer's size designation: _____
   c. Cross section code letter: _____
   d. Number of belts per drive: _____
   e. Horsepower transmitted per belt: _____
   f. Speed ratio: _____
   g. Pitch diameter of sheaves: Driver _____; driven _____
   h. Center to center distance of sheaves: _____
10. Flexible tubing:
    a. Manufacturer: _____
    b. Manufacturer's designation: _____
    c. Design pressure: _____ psig at _____ °F.
11. Pillow block:
    a. Manufacturer: _____
    b. Manufacturer's designation: _____

Figure 1. Pumping Assembly Data Sheet.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-P-23857B(YD) - PUMPING ASSEMBLIES, ASPHALT, BULK TRANSFER

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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